

Analysis of Metabolic Risk Factors

Using the NHANES 2013–2014 Dataset

Saci Nada - Mechou Lina

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Abstract

This study analyses metabolic risk factors using the NHANES 2013–2014 dataset ($n = 2118$), which contains demographic, physical, and blood-marker measurements for a sample of the US population. Two complementary objectives are pursued: (1) predicting insulin levels (LBXIN) from all available variables as a regression problem, and (2) classifying patients as presenting high abdominal obesity risk (waist ≥ 94 cm, WHO threshold) using blood markers and demographic variables with insulin as a continuous predictor. Linear regression, regression trees, and Random Forests are applied in Part 1; classification trees and Random Forests in Part 2. Together, the two parts form a coherent bilateral analysis confirming that the relationship between insulin and abdominal adiposity is bidirectional and robust across all modelling approaches.

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1 Introduction and Dataset

1.1 Dataset Description

The data come from the NHANES 2013–2014 public database. Three tables were merged on the unique participant identifier **SEQN**: *Age Prediction Subset*, *Body Measures*, and *Blood Pressure*. From the blood-pressure file only systolic (BPXSY1) and diastolic (BPXDI1) measurements were retained. The final dataset contains **13 variables** for **2118 individuals** with no missing values

Table 1: Variables available in the NHANES dataset

Variable	Type	Description
age_group	Categorical	Age group (Adult / Senior) — derived from RIDAGEYR
RIDAGEYR	Continuous	Age in years
RIAGENDR	Categorical	Sex (1 = Male, 2 = Female)
PAQ605	Categorical	Physical activity (1 = Yes, 2 = No, 7 = Refused)
BMXBMI	Continuous	Body Mass Index (kg/m ²)
LBXGLU	Continuous	Fasting glucose (mg/dL)
DIQ010	Categorical	Diabetes diagnosis (1 = Yes, 2 = No, 3 = Borderline)
LBXGLT	Continuous	2 h glucose OGTT (mg/dL)
LBXIN	Continuous	Insulin (μ U/mL) — regression target
BMXWAIST	Continuous	Waist circumference (cm) — classification target
LBDHDD	Continuous	HDL cholesterol (mg/dL)
BPXSY1	Continuous	Systolic blood pressure (mmHg)
BPXDI1	Continuous	Diastolic blood pressure (mmHg)

Categorical variables (RIAGENDR, DIQ010, PAQ605) were converted to factors in R. The dataset was split 70/30 into training and test sets:

```
sample <- sample(c(TRUE, FALSE), nrow(my_data_clean),
                 replace = TRUE, prob = c(0.7, 0.3))
train  <- my_data_clean[ sample, ]    # ~1481 observations
test   <- my_data_clean[!sample, ]    # ~637 observations
```

1.2 Study Objectives

- Regression (Part 1)** : Predict insulin levels (LBXIN) from all available variables. Three approaches are compared: linear regression, regression tree, and Random Forest.
- Classification (Part 2)** : Classify patients as *High* or *Normal* abdominal obesity risk using the WHO threshold of 94 cm for waist circumference. BMI is excluded because of its very high collinearity with waist , which would leak the answer to any classifier.

2 Part 1 Insulin Level Prediction (Regression)

2.1 Graphical Exploration

Insulin (LBXIN) is positively associated with BMI (BMXBMI) and waist circumference (BMXWAIST), suggesting that excess abdominal fat promotes hyperinsulinemia through

insulin resistance, further supported by the positive relationships with fasting glucose (LBXGLU) and the glucose tolerance test (LBXGLT). Conversely, HDL cholesterol (LBDHDD) shows a negative association with LBXIN, while blood pressure (BPXSY1, BPXDI1) exhibits only a weak relationship. Among categorical variables, diabetic status (DIQ010) appears to be the most discriminating factor, whereas sex and age group show no meaningful differences

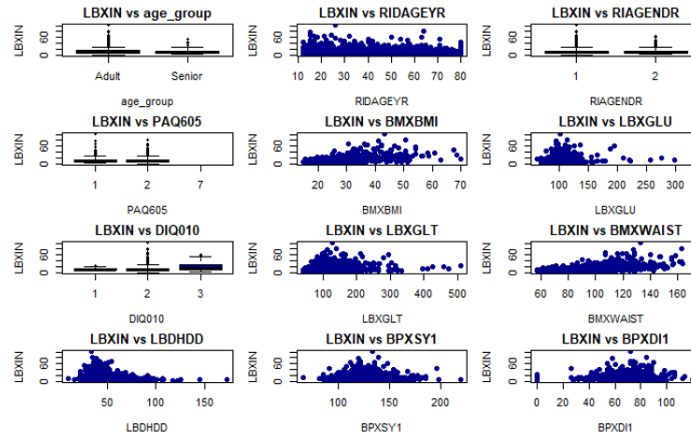


Figure 1: Graphical exploration: LBXIN vs. continuous predictors and categorical variables

2.2 Correlation Analysis

The correlation matrix confirms a very strong collinearity between BMI and waist. one of the two variables must be deleted.

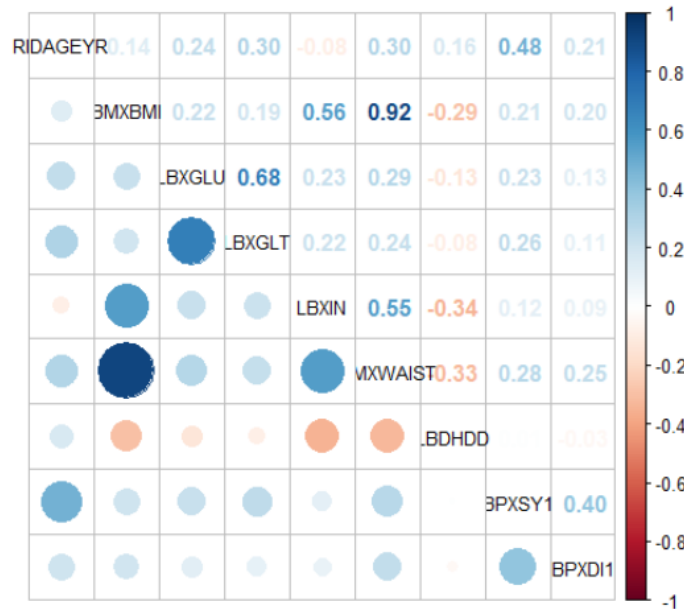


Figure 2: Correlation matrix

2.3 Linear Regression

Variable selection. Two models were compared: `model1` retains BMXBMI but excludes BMXWAIST; `model2` retains BMXWAIST but excludes BMXBMI. Model 2 achieves a

higher adjusted and is therefore selected. An exhaustive subset search (`regsubsets`, `method = "exhaustive"`) confirms that adjusted R^2 is higher when the subset contains `age_group`, `RIDAGEYR`, `RIAGENDR`, `DIQ010`, `LBXGLT`, `BMXWAIST`, `LBDHDD`, `BPXSY1`, `BPXD11` and `LBXGLU`

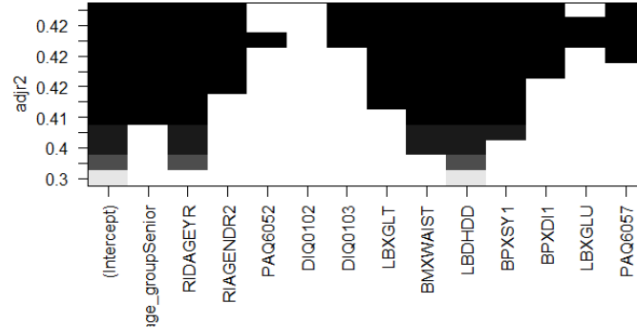


Figure 3: exhaustive search

Assumption checking and log-transformation. The model does not satisfy all linear regression assumptions, it violates linearity and homoscedasticity: residuals show non-constant variance and depart from normality (`check_model`). A **log-transformation** is applied to the target, which substantially improves residual behaviour (Figure 4).

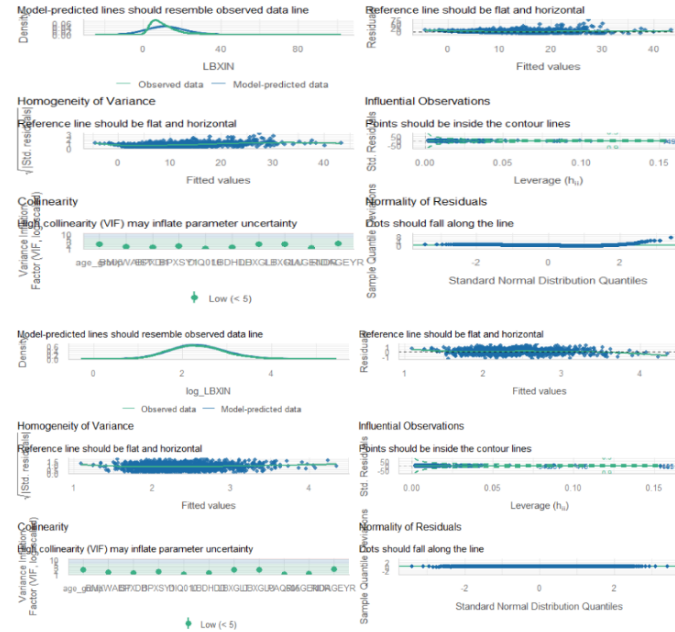


Figure 4: `check_model()` output before and after log-transformation of LBXIN

Final model. After log-transformation and standardisation of all numeric predictors, the final model includes: `age_group`, `RIDAGEYR`, `DIQ010`, `LBXGLT`, `BMXWAIST`, `LBDHDD`, `BPXSY1`, `BPXD11`, `LBXGLU`. Physical activity (`PAQ605`) is not significant and is dropped. The model achieves an adjusted $R^2 = 0.46$.

Test-set predictions. The model generally follows the trend, with better accuracy in the range (-1:1). However, it tends to overestimate low values and underestimate high values. The points furthest from the line correspond to atypical observations, explained by biological profiles not captured by the model or absent factors in the model.

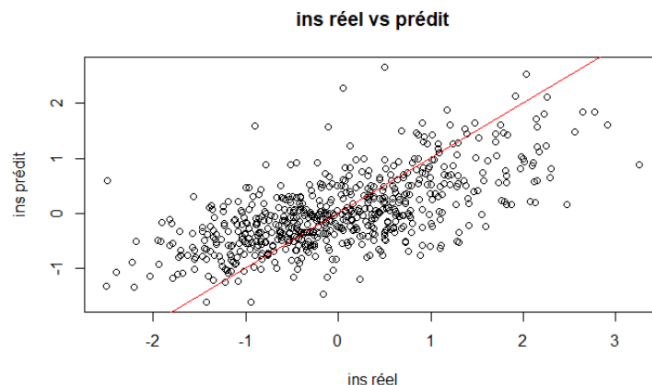


Figure 5: Predicted vs. actual

2.4 Regression Tree

A maximal unpruned tree ($cp = 0$) is grown on the training set, then the cross-validated relative error curve (`plotcp`) is used to select the optimal pruning threshold (Figure 6).

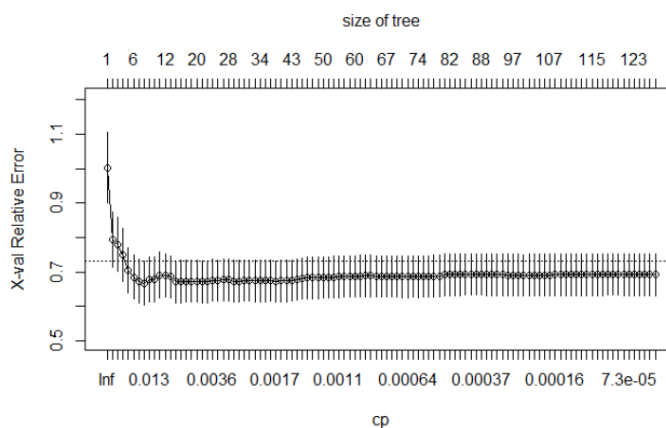


Figure 6: cp plot: maximal regression tree

Untransformed target. The pruned tree divides the population into 22 terminal nodes. Waist circumference (`BMXWAIST`) is the root split, confirming its dominant role in predicting insulin levels. The majority of individuals (84%) have a `BMXWAIST` < 110 cm, with a mean insulin level of $9.7 \mu\text{U/mL}$, and are further subdivided by BMI, age, fasting glucose, HDL and systolic blood pressure. For the remaining 16% (`BMXWAIST` ≥ 110 cm), insulin rises to $22 \mu\text{U/mL}$, with the highest values observed in individuals combining a large waist, low HDL and high blood glucose.

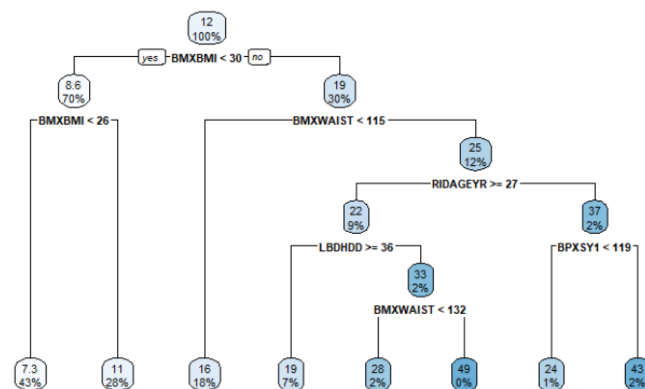


Figure 7: Pruned regression tree (`rpart.plot`) : untransformed LBXIN

After log-transformation. The cross-validated error drops, confirming that the transformation improves accuracy while yielding a simpler tree. BMXBMI replaces BMXWAIST as the root split on the log scale. Prediction plots show a clear improvement for extreme values (Figure 8).

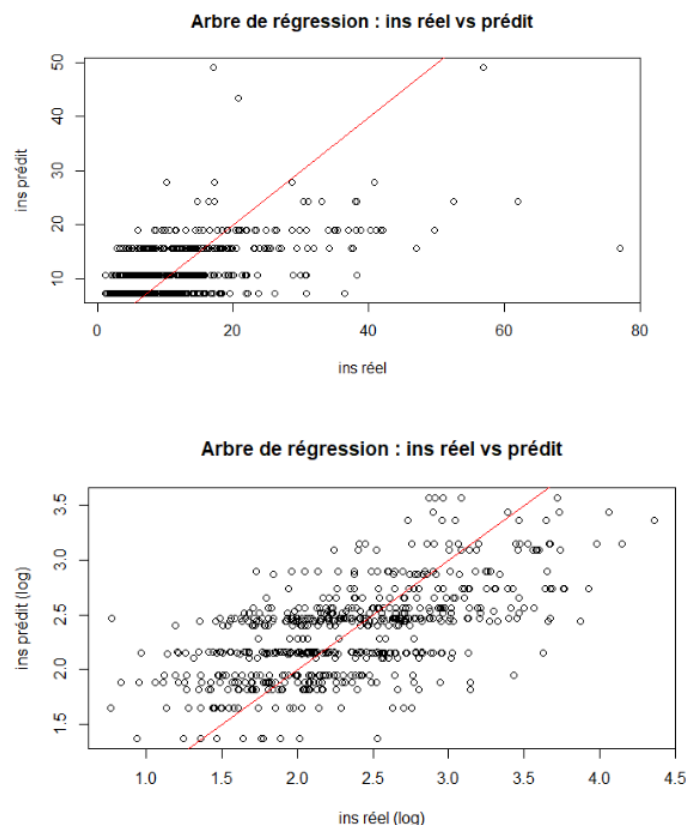


Figure 8: Predicted vs. actual insulin : regression tree, with and without log-transformation

2.5 Random Forest — Regression

A default Random Forest is first fitted to check error stabilisation, then the optimal `mtry` is selected by minimising the OOB error over a grid of 1 to 5 variables (Figures 9 and ??).

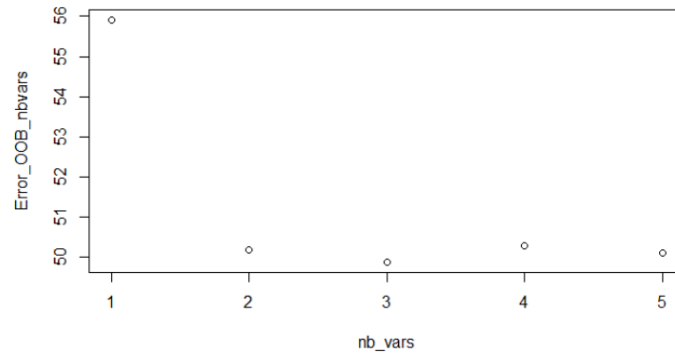


Figure 9: OOB error vs. nb vars

The OOB error is minimised at $mtry = 3$ (untransformed) and $mtry = 4$ (log-transformed); both final forests use 1000 trees. Without log-transformation, the Random Forest already follows the diagonal more closely than the single tree, but still underestimates high values. After log-transformation the improvement is striking: the points follow the diagonal consistently across all value ranges with notably more uniform dispersion (Figure 10).

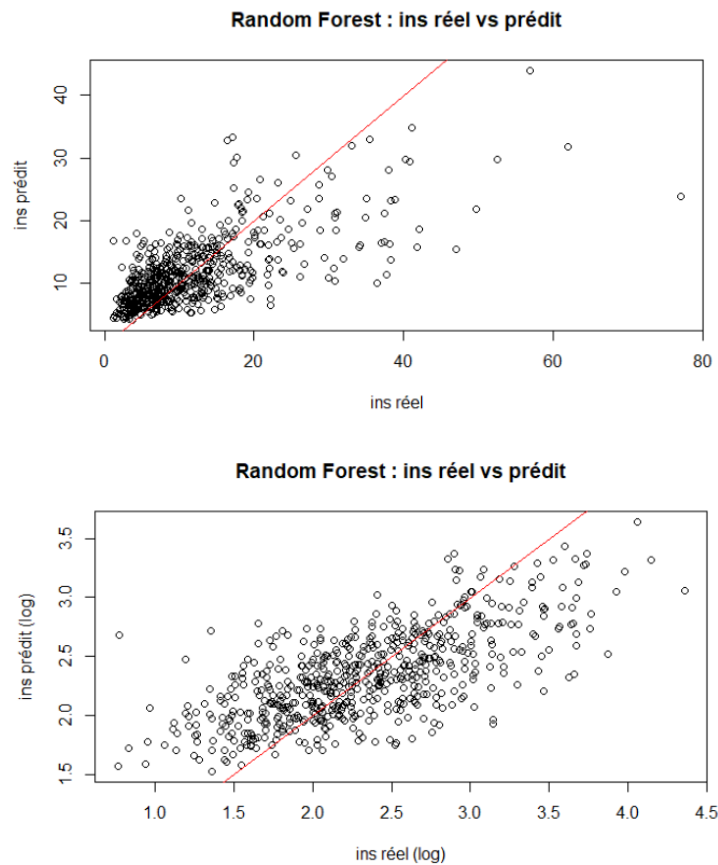


Figure 10: Predicted vs. actual insulin : Random Forest, with and without log-transformation

3 Part 2 Abdominal Obesity Classification

3.1 Data Preparation and Class Balance

A binary outcome **AbdominalObesity** is created using the WHO threshold of 94 cm: *High* (waist ≥ 94 cm) or *Normal* (waist < 94 cm). Classes are well balanced (49% High / 51% Normal), so no resampling correction is needed. BMI is removed from the feature set given its collinearity with waist ($r = 0.92$).

3.2 Dimensionality Reduction: PCA, t-SNE, UMAP

PCA. PC1 and PC2 together explain only 38.6% of variance, with substantial class overlap in the score plot. This signals that there is no simple linear boundary between *High* and *Normal* patients.

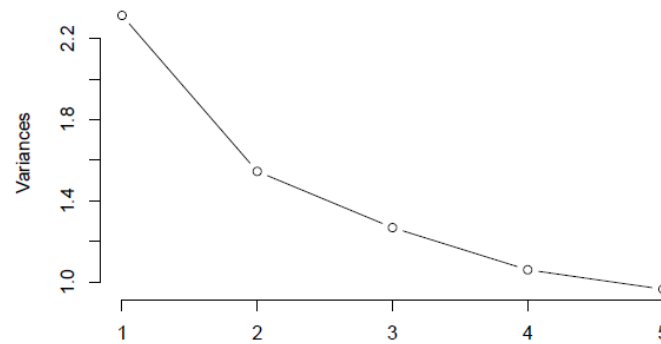


Figure 11: PCA score plot: class overlap between High and Normal groups

t-SNE and UMAP. The overlap in t-SNE and UMAP space suggests that many 'Normal' and 'High' patients share very similar metabolic profiles, which explains why the classification models struggle with high false-positive rates at a 50 percent threshold.

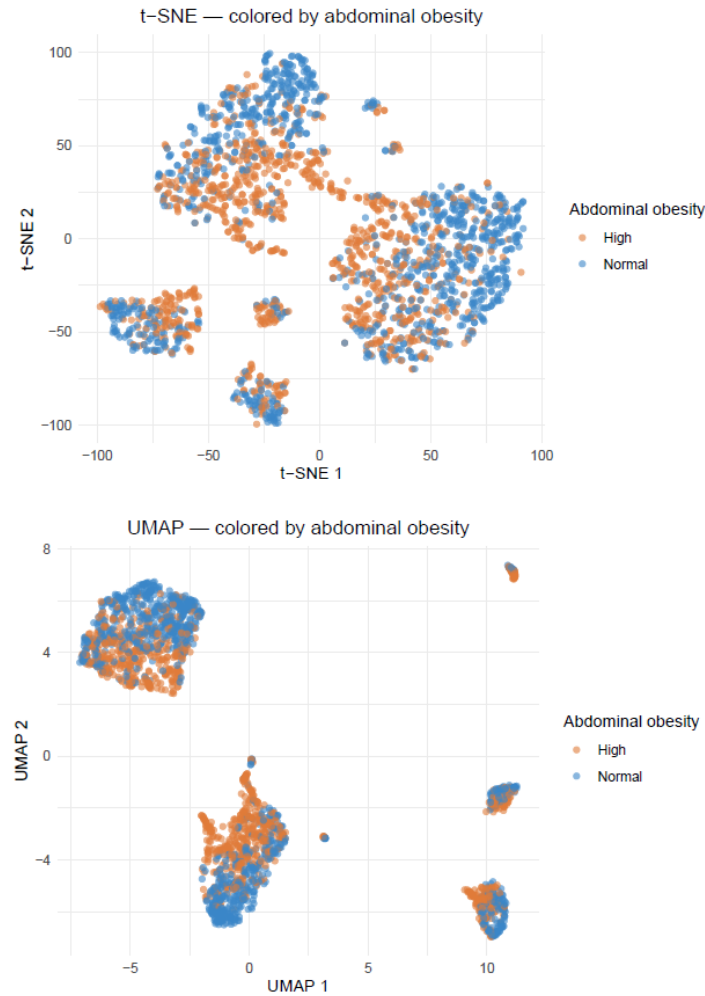


Figure 12: t-SNE and UMAP projections showing partial class separation

3.3 Random Forest Classification

Setup and tuning. A class weight of 1.5 is assigned to the *High* class to penalise the clinically costlier false negatives. The error curve stabilises well before 500 trees (Figure 13). The OOB error is minimised at `mtry = 2` over a grid of 1 to 10 variables (Figure 14); the final forest uses 200 trees.

```
optimal_forest <- randomForest(AbdominalObesity ~ ., data = train,
                               mtry = 2, ntree = 200,
                               importance = TRUE,
                               classwt = c("Normal" = 1, "High" = 1.5))
```

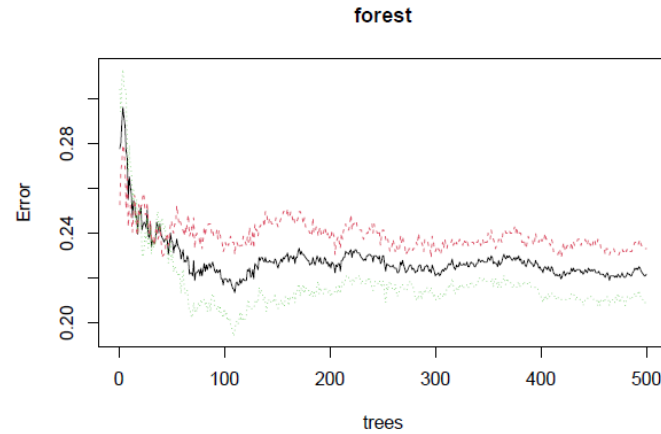


Figure 13: OOB error vs. number of trees : classification Random Forest

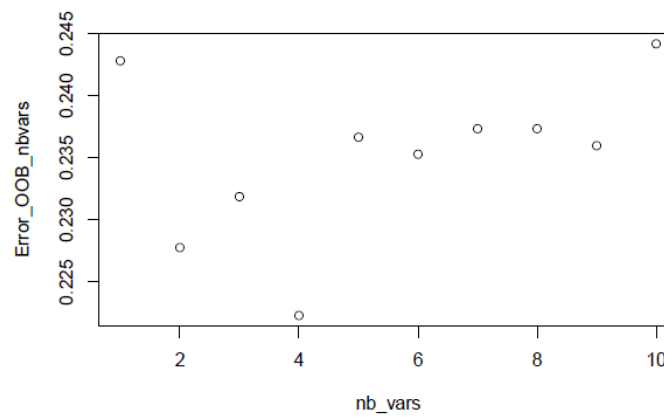


Figure 14: OOB error vs. mtry (grid 1–10) : classification Random Forest

Variable importance

The top three predictors by Mean Decrease in Accuracy are:

1. **Insulin (LBXIN)** by far the most important
2. **Age (RIDAGEYR)**
3. **HDL cholesterol (LBDHDD)**

This is consistent with the known metabolic syndrome cluster: central adiposity is tightly linked to insulin resistance and dyslipidaemia.

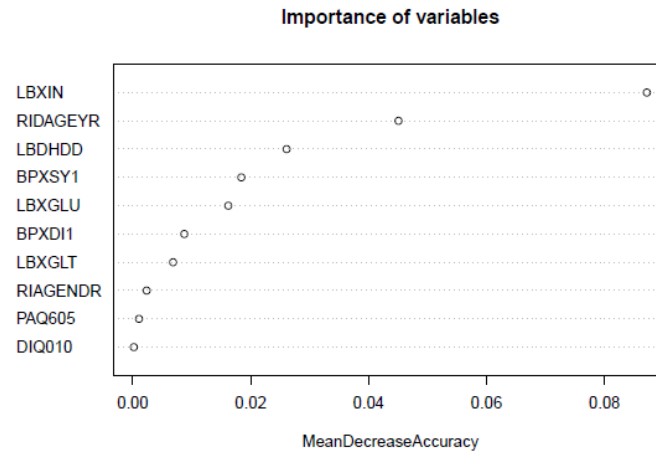


Figure 15: Variable importance plot (MeanDecreaseAccuracy) : classification Random Forest

Table 2: Random Forest classification performance on the test set

Metric	Value
Overall accuracy	77.6%
Sensitivity (<i>High</i> correctly identified)	75.6%
Specificity (<i>Normal</i> correctly identified)	79.4%
AUC (ROC)	0.859

Test-set performance. Sensitivity is slightly below specificity, reflecting the difficulty of detecting high-risk patients from blood markers alone, without any direct physical measurement (no waist or BMI in the feature set).

3.4 Classification Tree

A maximal tree ($cp = 0$) is grown, then pruned to the complexity parameter that minimises cross-validated error (Figure 16).

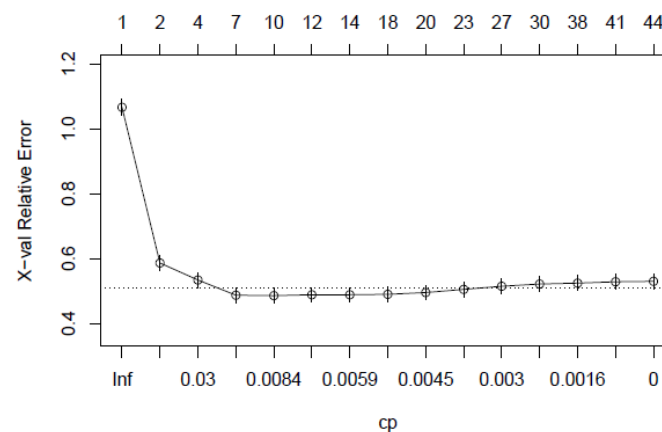


Figure 16: cp plot : maximal classification tree

The **first split of the pruned tree is insulin (LBXIN ≥ 11)**, confirming its dominant role. Subsequent splits involve age (RIDAGEYR), HDL cholesterol (LBDHDD), diastolic blood pressure (BPXD11), and OGTT glucose (LBXGLT), consistent with the Random Forest importance ranking. The tree is fully interpretable and provides an explicit clinical decision path.

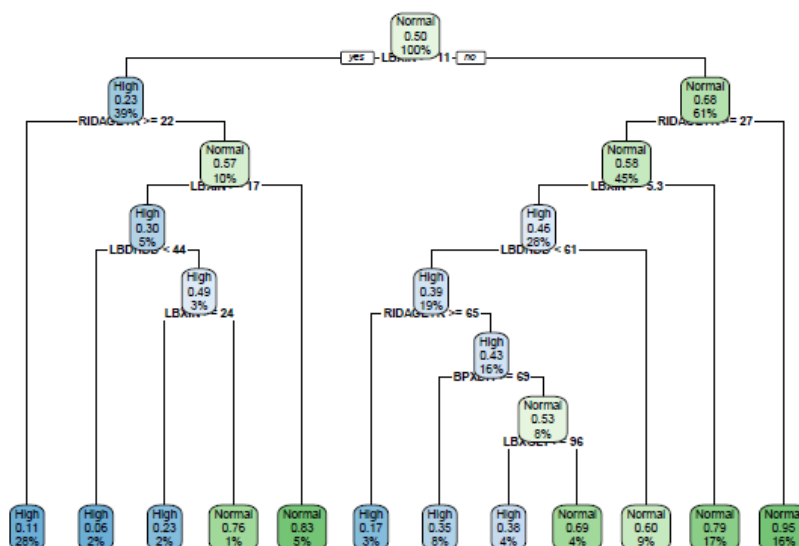


Figure 17: Pruned classification tree (rpart.plot)

Variable importance. Insulin, age, and HDL are again the top three predictors, consistent with the Random Forest results (Figure 18).

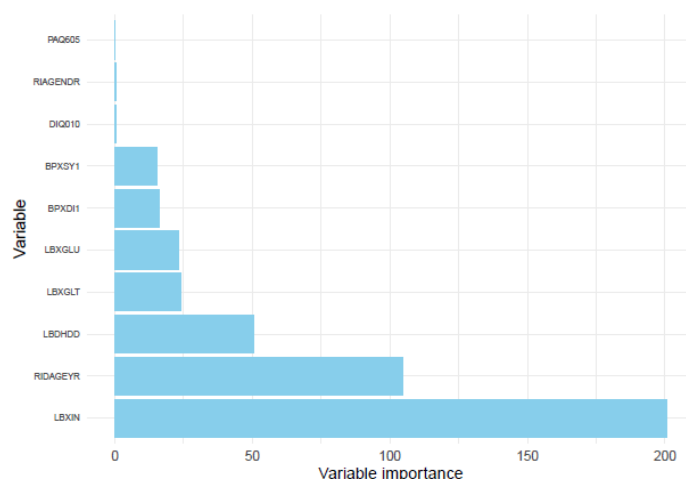


Figure 18: Variable importance bar chart : classification tree

Table 3: Classification tree performance on the test set

Metric	Value
Overall accuracy	76.4%
Sensitivity (<i>High</i>)	73.7%
Specificity (<i>Normal</i>)	78.8%
AUC (ROC)	0.830

Test-set performance. The classification tree performs slightly worse than the Random Forest, which is expected: a single tree has higher variance. It remains interpretable and gives comparable results, making it a useful clinical complement.

3.5 ROC Curve Comparison

Both models perform well above chance across all decision thresholds. The Random Forest (AUC = 0.859) outperforms the classification tree (AUC = 0.830), with a better overall sensitivity/specificity trade-off (Figure 19).

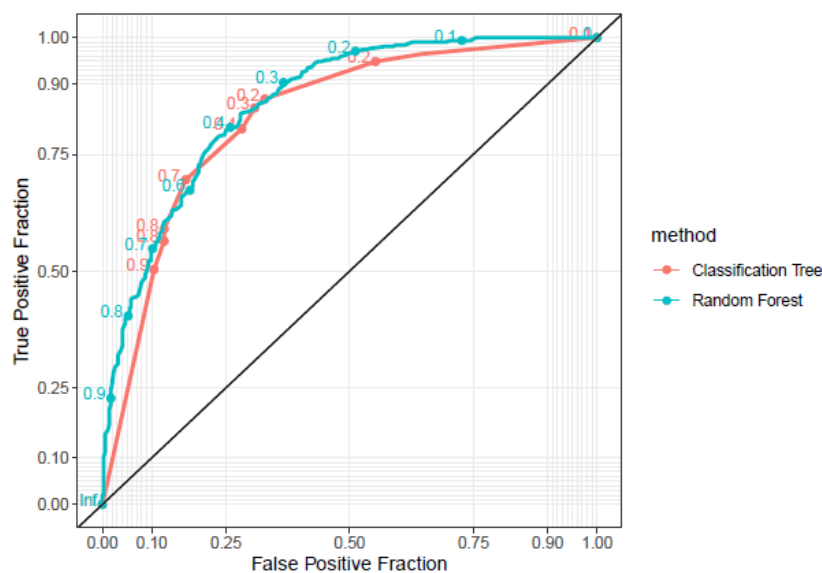


Figure 19: ROC curve comparison: Random Forest vs. Classification Tree

Table 4: Classification model summary

Model	Accuracy	Sensitivity	Specificity	AUC
Random Forest	77.6%	75.6%	79.4%	0.859
Classification Tree	76.4%	73.7%	78.8%	0.830

4 Discussion and Conclusion

4.1 Common Findings in Both Parts

The most important result that appears in both parts of the study is the key role of insulin. In the regression part, insulin levels are best predicted by waist circumference and BMI. In the classification part, insulin is the strongest predictor of high abdominal obesity risk. This goes both ways: having too much abdominal fat leads to higher insulin levels, and high insulin levels are a sign of abdominal obesity. This pattern is well known in the context of metabolic syndrome and is confirmed here across all the models we tested.

Three variables consistently come up as the most important across all models: waist circumference and BMI (which are very strongly correlated and basically measure the same thing), age (RIDAGEYR), and HDL cholesterol (LBDHDD).

4.2 Model Comparison

For the regression part, the Random Forest applied on the log-transformed insulin gives the best prediction results. The linear model is less accurate but easier to interpret since it clearly shows the effect of each variable. All models struggle with very high insulin values, which is probably because some important factors like medication or genetic background are not included in the dataset.

For the classification part, the Random Forest performs better with an AUC of 0.859 compared to 0.830 for the classification tree. However the classification tree is easier to read and understand, and gives a clear step by step decision path that could be useful in a medical context.

4.3 Limitations

- Insulin values are not evenly distributed and are skewed to the right, which makes it hard to predict very high values well, even after applying the log-transformation.
- The 94 cm waist cutoff is a general population threshold and may not be the best criterion for every individual case.
- The data were collected at a single point in time, so we cannot draw any conclusions about cause and effect.
- Physical activity (PAQ605) turned out to have very little impact on predictions across all models, which may be because it was self-reported and therefore not very accurate.

4.4 Conclusion

This study shows that insulin levels and abdominal obesity risk can be predicted with reasonable accuracy using a small set of blood markers and demographic variables. The main findings are:

1. Waist circumference is the most important predictor of insulin levels across all regression models, followed by age and HDL cholesterol.

2. Insulin is the most important predictor of abdominal obesity in both the Random Forest and the classification tree, which confirms that the relationship between the two goes both ways.
3. Applying a log-transformation to insulin improves the accuracy of all regression models, reducing the cross-validated error from around 0.68 to around 0.60 for the trees.
4. Random Forests give better results than single decision trees in both parts of the study, with an AUC of 0.859 compared to 0.830 for classification.
5. The connection between insulin and abdominal fat is consistent across all models and methods tested in this study.